

# Rohan Siva

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## EDUCATION

### University of Texas at Austin

**Expected: May 2027**

*B.S. Electrical and Computer Engineering Honors, Minor in Stats & Data Science*

*GPA: 3.98/4.0*

**Relevant Coursework:** Data Structures, Algorithms, Machine Learning, Data Science Principles, Data Science Lab, Probability and Statistics, Object-Oriented Design, Discrete Math, Multivariable Calculus, Linear Algebra, Differential Equations, Embedded Systems

## SELECTED PUBLICATIONS

**Siva, R.,** Bhatt, N. P., Yang, Y., Lee, S., Gadde, N., Ellis, C., Velasquez, A., Wang, Z., & Topcu, U. "What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning," In Proceedings, **Conference on Robot Learning (CoRL), 2026.**

Bhatt, N. P., Yang, Y., **Siva, R.,** Samineni, P., Milan, D., Wang, Z., & Topcu, U. "VLN-Zero: Rapid Exploration and Cache-Enabled Neurosymbolic Vision-Language Planning for Zero-Shot Transfer in Robot Navigation," In Proceedings, **NeuS, 2026. (Spotlight)**

Bhatt, N. P., Yang, Y., **Siva, R.,** Milan, D., Topcu, U., & Wang, Z. "Know Where You're Uncertain When Planning with Multimodal Foundation Models: A Formal Framework," In Proceedings, Conference on Machine Learning and Systems (**MLSys**), **2025. (Oral)**

## EXPERIENCE

### Decagon | *Research Intern* | San Francisco, CA

*June 2026 - Present*

- Architected voice inference engine using continuous batching and paged KV caching; achieved 27ms latency and 10x throughput
- Designed novel RL algo for flow-matching speech models to optimize voice naturalness and control tags, with 98.2% stability
- Shipped audio signal-to-noise ratio gate, filtering ~10% of noisy audio segments, and improving turn detection by 15%

### UT Austin Center for Autonomy | *Researcher* | Austin, TX

*Aug 2024 – Present*

- Co-authored 8 papers on autonomous planning, perception, and computer vision: 2 oral presentations and Best Paper Nomination
- Developed latent-space failure prediction and steering for  $\pi 0.5$  VLAs, evaluated on 1,151 LIBERO rollouts with 91.5% success
- Built function-based robot learning methods achieving 25–36% gains over VLM baselines, and 100–350× faster inference

### Mercor | *Machine Learning Engineer Intern* | Remote

*Aug 2025 - Dec 2025*

- Architected end-to-end LLM benchmarking pipeline for OpenAI MLE tasks using Retrieval-Augmented Generation (RAG) and PyTorch, benchmarking over 12,000+ OpenAI MLE tasks/month and reducing manual review time by 85%
- Overhauled LLM evaluation pipeline, with sub-200ms latency and cut compute costs by 30% via optimized embedding caching

### Statistical Learning & AI Group Lab @ UT Austin | *Researcher*

*May 2024 – Aug 2024*

- Optimized speculative decoding for GRPO, increasing acceptance rate by 15% via training trajectory-conditioned EAGLE drafter
- Explored custom GRPO and PPO implementations for NanoChat and improved post-RL GSM8K performance by ~12%

### Cisco Systems | *Machine Learning Engineer Intern* | San Francisco, CA

*May 2025 – Aug 2025*

- Created an agent to convert natural language into ETL pipelines with direct enterprise PostgreSQL data loading, adopted by 9+ internal teams, decreasing pipeline-creation tickets by 80% per quarter and reducing setup time by 70%. Published as a paper.
- Built full-stack microservice with React, Elasticsearch, Bedrock, and AWS Lambda for deployment, delivering 99.7% uptime
- Eliminated release failures with CI/CD for Docker and Kubernetes deployments and unit tests, achieving 90%+ coverage

## PROJECTS

### MoodScribe: Emotion-based Chatbot | [Github](#)

*Nov 2024*

- Architected a full-stack web app with CyberPunk-themed UI (HTML/CSS) to generate sentiment-based Instagram reply suggestions, implementing async Meta API scraping, caching, and FastAPI endpoints to serve 2500+ concurrent users
- Fine-tuned Llama 3.1 using Unsloth and quantized LoRA adapters for sentiment analysis, cutting inference latency by 44%

### NutriCheck: Image Classification for Nutrition | [Github](#)

*Oct 2024*

- Built an image classification app with custom Keras CNNs for real-time food analysis, retrieving nutritional data from Supabase
- Fine-tuned MobileNetV2 to classify food and display top 3 predictions with 92% accuracy, and output key nutritional content

### Minerva: AI-Powered Study App | [Github](#)

*Dec 2024*

- Engineered a Flask and React AI study app for 300+ UT students with OpenAI API for tailored quizzes, flashcards and chatbot
- Developed a Retrieval-Augmented Generation pipeline using LlamaIndex to retrieve study material via Google Classroom API

## SKILLS & INTERESTS

**Languages:** Python, Java, C/C++, JavaScript, SQL, TypeScript, HTML/CSS, Swift

**Frameworks & Tools:** React.js, Node.js, Flask, Docker, Kubernetes, PostgreSQL, AWS, Git, CI/CD, Azure, GCP, NumPy, Pandas

**AI/ML:** Pytorch, Keras, Sklearn, OpenCV, LangChain, Transformers, Apache Spark, Apache Kafka